

Kamal Manchenella

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Darkxzie

Education

- 2024 – **B.E. in Mathematics and Computing**, *BITS Pilani, Goa Campus*, India
Present Currently in 2nd Year
- 2022 – 2024 **Senior Secondary Education (Class XII)**, *Delhi Public School, Hyderabad*, India
Graduated with a focus on Science and Mathematics

Work Experience

- Apr 2026 – **Research Intern**, *Deep Forest Sciences*
Present Working on DeepRetro, a hybrid retrosynthesis planning framework integrating machine learning and rule-based methods for automated reaction pathway generation.
- Dec 2025 – **Student Researcher**, *APPCAIR (Applied AI Research Center), BITS Pilani*
Present Working under Prof. **Tanmay Tulsidas Verlekar** on peer-review rebuttal alignment in collaboration with **TCS Research**
- Dec 2024 – **Software Intern**, *SpotWays*
Jan 2025 Developed a stop detection algorithm using GPS coordinates and timestamps.
- Aug 2025 – **Student Researcher**, *Pragya*
Nov 2025 Contributed to ongoing research in Large Language Models under Prof. **Amitava Das** through the internship program.

Volunteering

- Aug 2025 – **Campus Partner**, *Perplexity*
Jan 2025 Supporting Perplexity's campus initiatives as a student partner.
- Aug 2025 – **Instructor**, *Center for Technical Education, BITS Goa*
Jan 2025 Taught an intermediate course on *Building LLMs from Scratch*, covering transformers, training pipelines, fine-tuning, and optimization of modern large language models.

Projects

- Robust Loss Functions** Explored and implemented CE, FL, NCE, NFL, and APL loss functions under varying noise rates. Built CNNs from scratch and evaluated model performance across 40% noise-injected CIFAR-10 data.
- Diffusion** Implemented a diffusion-based generative model (DDPM) trained on CIFAR-10. Investigated the impact of CFG, sampling steps, and attention mechanisms (DiT vs Xformers) on generation quality using FID and CMMD scores.
- Obesity Risk Prediction** Implemented and compared ensemble methods (Random Forest, Gradient Boosting, XGBoost) on structured health data to classify individuals into obesity categories. Achieved 96% accuracy using a voting classifier.

Neural Net from Scratch Built a fully connected neural network using only NumPy to classify handwritten digits from the MNIST dataset. Trained with SGD, reaching 97.8% accuracy.

Technical Reports

- Apr 2025 **ML and Diffusion Report** – Authored detailed analysis of custom loss functions for noisy CIFAR-10 classification and diffusion model performance. Covered model architectures, evaluation under label noise, and metrics like FID and CMMD.
- Jun 2025 **Ensemble Learning and Neural Net Report** – Wrote an in-depth report covering ensemble learning for obesity risk prediction and neural network implementation for MNIST. Emphasized performance comparisons and hands-on architecture design.

Clubs & Societies

- Jun 2025 – **Member**, *Society for Artificial Intelligence and Deep Learning (SAIDL)*, Goa, India
Present Involved in machine learning-based research and peer collaboration.
- Aug 2024 – **Marketing Associate**, *Dept. of Sponsorship and Marketing, BITS Goa*, Goa, India
Aug 2025 Responsible for cold calling, sponsor outreach, and promotional strategy.
- Aug 2024 – **Junior Developer**, *Developers' Society, BITS Goa*, Remote
Sep 2025 Working with C++ and Python on society projects and internal tooling.

Skills

- Languages C, C++, Python, Java
Tools Git, GitHub, LaTeX
Other Event Planning, Marketing, Remote Collaboration

Certifications

- Mar–Sep 2025 **Machine Learning Specialization**, DeepLearning.AI – Stanford University — Covered supervised, unsupervised, and reinforcement learning; model evaluation; and practical ML system design.
- Aug–Sep 2025 **Deep Learning Specialization**, DeepLearning.AI — Covered neural networks, CNNs, RNNs, sequence models, hyperparameter tuning, and model optimization.